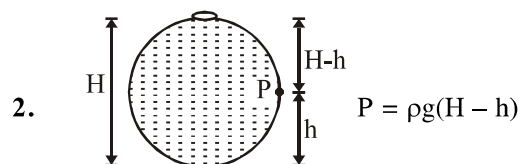


Solution :-

1. $V = V_1 + V_2$

$$\frac{2m}{\rho_{\text{mix}}} = \frac{m}{\rho_{\omega}} + \frac{m}{2\rho_{\omega}} \Rightarrow \frac{2m}{\rho_{\text{mix}}} = \frac{3m}{2\rho_{\omega}}$$

$$\rho_{\text{mix}} = \frac{4\rho_{\omega}}{3} \Rightarrow \frac{\rho_{\text{mix}}}{\rho_{\omega}} = \frac{4}{3}$$



3. Due to decrease in outside pressure volume of the bubble increases hence radius also increases.

4. Upthrust $F_{\text{th}} = \rho V_{\text{in}} g_{\text{eff}} = 0$, $g_{\text{eff}} = 0$
[Freely falling body]

5. $F_{\text{th}} = \rho_l V_{\text{in}} g$
Force due to liquid depends on density of liquid, volume of liquid displaced by the body and acceleration due to gravity.

6. As the velocities at cross sections of the horizontal pipe connected with tubes A and C are same therefore by Bernoulli's theorem pressure at these cross sections are also same, so height of liquid in tubes A and C are same.

7. Excess pressure inside soap bubble is inversely proportional to, the radius of bubble.

i.e. $\Delta P \propto \frac{1}{r}$

So A and C have more inside pressure than that of B. So the air will move from A and C towards B.

8. (Rate of flow) = $\frac{\pi \Delta P r^4}{8\eta l}$

$$V \propto \frac{r^4}{l} \Rightarrow \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^4 \left(\frac{l_2}{l_1}\right)$$

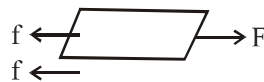
$$\Rightarrow \frac{V_1}{V_2} = \left(\frac{D/2}{D/4}\right)^4 \left(\frac{2l}{l}\right) = 32$$

$$\therefore V_2 = \frac{V_1}{32} = \frac{V}{32}$$

for 4 pipes

$$V_2' = 4V_2 = 4 \times \frac{V}{32} = \frac{V}{8}$$

9. $f = \eta A \frac{dv}{dy}$



$$F = 2f \text{ (as net force should be zero)}$$

$$\Rightarrow F = 2\eta A \frac{dv}{dy}$$

$$\Rightarrow 1 = 2\eta \times 0.5 \times \frac{0.5}{1.25 \times 10^{-3}}$$

$$\eta = \frac{1.25 \times 10^{-3}}{0.5} = 2.5 \times 10^{-3} \text{ kg/m-sec.}$$

10. $v_r = \frac{2}{9} r^2 \frac{g(\rho - \sigma)}{\eta}$

$$\frac{v_2}{v_1} = \left(\frac{r_2}{r_1}\right)^2 \Rightarrow v_2 = \left(\frac{0.3}{0.2}\right)^2 (4) = 9 \text{ cm/s}$$

12. $h\rho g = \frac{2T}{R}$

$$h \downarrow \Rightarrow R \uparrow$$

13. $P_{\text{ex}} = \frac{2T}{r}$

$$\text{if } r \uparrow \Rightarrow P_{\text{ex}} \downarrow$$

$$V_T = \frac{2}{9} r^2 \frac{g(\rho - \sigma)}{\eta}$$

$$V_T \propto r^2$$

$$\text{if } r \uparrow \Rightarrow V_T \uparrow$$

14. $F = \left(\frac{Y_A}{L}\right) \Delta l = kx$

$$\therefore k = \frac{Y_A}{L}$$

15. Characteristics of fluids are

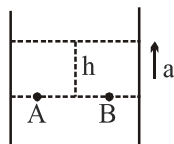
(i) The atoms or molecules in a fluid are arranged in a random manner.

(ii) A fluid cannot withstand tangential or shearing stress for an indefinite period. It begins to flow when a shearing stress is applied.

(iii) A fluid has no definite shape of its own. It ultimately takes the shape of the containing vessel.

16. Pressure is same at all point which are at same height.
17. Transmitted undiminished.
19. The fluid does not have any resistance to change off its shape. Thus, the shape of a fluid is governed by the shape of container. A fluid offers resistance to change in pressure and volume defined by its bulk modulus.
20. $P = P_0 + \rho gh$
depends on ρ (density) g (acceleration due to gravity) h (depth from free surface)
21. Most viscous fluid comes to rest earlier due to dissipation of energy at a larger rate.
22. When temperature of liquid increases, its viscosity decreases.
23. In a streamline flow velocities of all the particles passing through a given point are same.
24. Terminal velocity
- $$v_T = \frac{2}{9} \frac{r^2 g (\rho_b - \rho_l)}{\eta}$$
- $$\Rightarrow v_T \propto \rho_b - \rho_l$$
26. In gases, as the temperature increases, the random motion of the atoms increases so viscosity of gases also increases.
Viscosity of gases are practically independent of pressure.
27. Viscous force on the body by the fluid is
 $F = 6\pi\eta r v$
28. Surface tension of a liquid becomes zero at its boiling temperature.
29. When equal volumes are poured in the vessel then maximum height of liquid will be in the vessel C.
30. When equal masses of 3 liquids are kept in identical vessels then weight of all 3 liquids are same.
31. Pressure due to air on liquid decreases therefore the total pressure at the bottom of the liquid also decreases.

32.



$$p_A = p_B = p_0 + \rho g_{\text{eff}} h \quad p_A > p_B$$

$$= p_0 + \rho (g + a) h$$

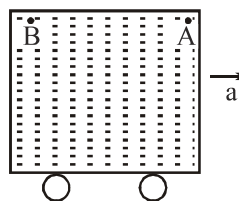
33. $p_0 = \rho(g + a)(76 \text{ cm})$ (when lift is accelerated upward).

$$p_0 = \rho gh \text{ (when lift is at rest)}$$

$$(h > 76 \text{ cm})$$

34. Pushing the air inside bottle will increase its density, hence the net buoyant force on the block due to air. So lesser part will be in water.

35.



$$p_B > p_A$$

36. At same horizontal level :

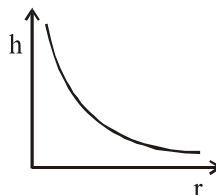
$$Av = \text{constant. ... (i)}$$

$$p + \frac{1}{2} \rho v^2 = \text{const. ... (ii)}$$

37. Due to surface tension liquid takes a shape which have lesser surface area.

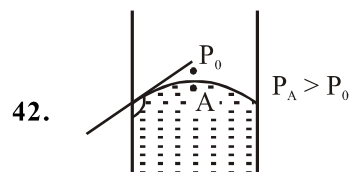
38. When drops merge than surface area decreases. That's why surface energy also decreases.

39. $h \propto \frac{1}{r}$ (Zurin's law)

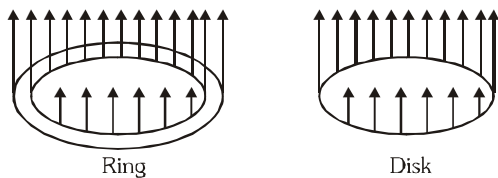


40. Viscosity is the property of fluids i.e. liquids and gases.

41. Liquid does not rise in the capillary if -
Either (i) its surface tension is zero.
or (ii) angle of contact is 90.



43. Ring has double surface than that of disk.



44. Temperature $\uparrow$ surface tension $\downarrow$

45. $6\pi\sigma r v_t = \frac{4}{3}\pi r^3 (\rho - \sigma) \Rightarrow v_t \propto r^2$

48. $RD = \frac{\text{weight in air}}{\text{loss of weight in water}} = \frac{5.00}{1.00} = 5.00$

$$\frac{d\rho}{\rho} = \frac{0.05}{5.00} + \frac{0.1}{1.00} = 0.11 \text{ or } 11\%$$

$$\therefore \rho = 5.00 \pm 11\%$$

49. After proportionality limit, strain rise is relatively more.
50. Aluminum rod will try to expand faster.